

VULNERABILITY ASSESSMENT REPORT
REMOTE DATABASE SERVER – PUBLIC ACCESSIBILITY RISK
PREPARED BY: CYBERSECURITY ANALYST | DATE: MARCH 25, 2026
FRAMEWORK REFERENCE: NIST SP 800-30 REV. 1

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The purpose of this vulnerability analysis is to evaluate the risks associated with the publicly accessible database server that stores critical business and customer information. This server is highly valuable because it supports daily operations, including identifying potential customers and enabling remote employee access. Securing this data is essential to prevent unauthorized access, data breaches, and potential data manipulation. If the server were compromised or disabled, it could disrupt business operations, damage customer trust, and result in financial and reputational losses.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
External hacker	Obtain sensitive information via data exfiltration	3	3	9
Privileged user	Alter or delete critical information	2	3	6
Technological threat	Conduct a Denial of Service (DoS) attack	3	2	6

Approach

The selected threat sources and events were chosen because they directly relate to the database server being publicly accessible, which significantly increases exposure to both external and internal risks. External hackers and DoS attacks were included due to their high likelihood and common occurrence against internet-facing systems. Insider threats were considered because privileged users have direct access to sensitive data and can cause severe damage, whether intentional or accidental. These risks are significant because they can lead to data breaches, service disruptions, and loss of business continuity, ultimately impacting customer trust and organizational reputation.

Remediation Strategy

To remediate the identified risks, the organization should restrict public access to the database server and implement the **principle of least privilege**, ensuring users only have access to the data necessary for their roles.

Multi-factor authentication (MFA) should be enforced to prevent unauthorized access, especially for privileged accounts.

A **defense-in-depth strategy**, including firewalls, intrusion detection systems, and regular patching of Linux and MySQL, can help reduce exposure to external attacks.

Additionally, implementing an **Authentication, Authorization, and Accounting (AAA)** framework will allow proper monitoring, access control, and logging of user activities to detect and respond to suspicious behavior.